

Introducing metal–sulfur active sites in metal–organic frameworks via post-synthetic modification for hydrogenation catalysis

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Metal–sulfur active sites play a central role in catalytic processes such as hydrogenation and dehydrogenation, yet the majority of active sites in these compounds reside on the surfaces and edges of catalyst particles, limiting overall efficiency. Here we present a strategy to embed metal–sulfur active sites into metal–organic frameworks (MOFs) by converting bridging or terminal chloride ligands into hydroxide and subsequently into sulfide groups through post-synthetic modification. We apply this method to two representative MOF families: one featuring one-dimensional metal–chloride chains and another containing discrete multinuclear metal clusters. Crystallographic and spectroscopic analyses confirm structural integrity and sulfide incorporation, and the transformation is monitored by in situ total scattering methods. The sulfided MOFs display enhanced catalytic activity in the selective hydrogenation of nitroarenes using molecular hydrogen. Density functional theory calculations indicate that sulfur incorporation promotes homolytic metal–ligand bond cleavage and facilitates H₂ activation. This work establishes an approach to construct MOFs featuring accessible metal–sulfide sites.

Highly polarizable components, such as sulfur-containing groups, in catalytic processes have emerged as a forefront area of research due to their unique ability to modulate chemical reactivity through electronic coupling. In this context, metal–sulfide compounds are of particular interest as they leverage the electronic flexibility of sulfur centres^{1–5}, resulting in a more covalent nature of M–S bonds compared with their oxygen analogues, enhancing the charge-transfer properties that play important roles in catalytic processes^{1,6–9}. Nature’s ingenious use of

sulfur-containing species, such as Fe₄S₄, Fe–V–S, Fe–Mo–S clusters and Cu–S moieties, in the active sites of enzymes for critical redox reactions has afforded excellent reactivity within biological systems for transformations such as hydrogenation and dehydrogenation^{8,10–15}. Despite their efficacy, however, the fragility and complexity of enzyme systems often limit their scalability and industrial application¹⁶.

In contrast to enzymatic sulfur-containing species, transition metal–sulfur compounds, exemplified by MoS₂ (refs. 17–20) and CoS_x

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(refs. 21–23), offer relatively greater stability and have been demonstrated to play a essential role in various catalytic processes, including hydrogen splitting and hydrogenation^{6,7,21,22}. The catalytic efficacy of these compounds is intrinsically tied to the local environment of the active sites, where the morphology of particles, particularly at the surfaces and edges, becomes crucial for catalytic performance. This necessitates sophisticated defect engineering to maximize exposure of active sites. It has been demonstrated that increasing the surface areas of these systems improves the catalytic performance as the active sites are more easily accessible, ensuring efficient mass transport^{24–27}.

Beyond these well-known materials, metal–organic frameworks (MOFs) represent a burgeoning class of materials with notable implications for catalysis. The structural robustness and high internal surface areas of MOFs make them ideal for serving as high-performing catalysts and hosting catalytic substrates^{28–30}. In fact, encapsulating enzymes within MOF pores can substantially enhance both stability and catalytic efficiency relative to those of the free enzyme^{31–34}. Furthermore, MOFs featuring open metal sites as reactive centres have been identified as promising candidates in catalysis, offering well-organized and protected reactive centres in a solid-state framework^{30,35–37}. However, the synthesis of MOFs incorporating highly polarizable components, such as sulfide bridges, metal–sulfur-based clusters or thiolate linkers, remains largely underexplored, with only a few known examples^{38–45}. The primary challenges hindering progress in this area include the instability of metal–sulfur clusters, the comparatively lower crystallinity than metal–oxide-based MOFs resulting from the irreversible nature of forming highly polarized bonds, the poorly understood effects of templating agents on sulfur-containing structures, and the overarching need for novel synthetic methodologies to achieve desired topologies³⁸.

Post-synthetic modification⁴⁶ has emerged as a promising strategy to modify bridging linkers and/or transform metal clusters, such as metal–oxide clusters, into their sulfur counterparts by leveraging the inherent topology of parent MOFs, the extensive database of known structures and the simplified synthetic conditions required (Fig. 1a). This approach not only mitigates the challenges associated with direct synthesis but also opens new avenues for the development of advanced catalytic materials.

In this study, we selected two families of specific MOFs with potentially redox-active metal centres and accessible open metal sites: M_2Cl_2BBTA ($M = Co, Ni$; $H_2BBTA = 1H,5H$ -benzo(1,2-d:4,5-d')bistriazole; Fig. 1b)^{47–49} and $M-MFU-4l-Cl$ ($M_xZn_{5-x}Cl_4(BTDD)_3$; $x = 4$ for $M = Co$ and $x = 2.5$ for $M = Ni$; $H_2BTDD = bis(1H-1,2,3-triazolo[4,5-b][4',5'-i])dibenzo[1,4]dioxin$; Fig. 1c)^{50–52} as parent MOFs. We converted them to their hydroxide-bridged versions, $M_2(OH)_2BBTA$ and $M-MFU-4l-OH$, and subsequently to their sulfided versions, $M_2(SH)_2BBTA$ and $M-MFU-4l-SH$, and elucidated their crystal structures using a combination of single-crystal X-ray diffraction and electron diffraction techniques. In situ pair distribution function (PDF) analysis revealed real-time insights into the sulfidation process, enabling dynamic tracking of structural transformations during modification. We evaluated the catalytic performance of the sulfided MOFs with a model reaction using chemoselective hydrogenation of nitroarenes—the reduction of aryl–NO₂ to –NH₂. The sulfided MOFs showed markedly enhanced hydrogenation reactivity compared with their parent MOFs, which contain terminal or bridging chloride and hydroxyl groups. Density functional theory (DFT) calculations clarified the mechanism, showing sulfide ligands promote the homolytic cleavage of metal–ligand ($M-X$) bonds, leading to more efficient H₂ activation. These integrated experimental and computational investigations reveal key reactivity trends and emphasize the crucial role sulfur incorporation plays in modulating the catalytic properties of MOFs.

Results and discussion

Sulfidation of MOFs via post-synthetic methods

For the post-synthetic modification, we selected two types of MOFs, M_2Cl_2BBTA and $M-MFU-4l-Cl$ ($M = Co, Ni$), to investigate how isolated

metal nodes or one-dimensional (1D) chains with adjacent centres affect the properties of hydrogenation reactions. Both MOFs have similar triazolate linkers as supporting ligands and comparable pore sizes (Fig. 1). In enzymes, nitrogen-containing heterocyclic groups such as imidazoles have been proposed to extend the conjugation system for electron tunnelling with metal–sulfide clusters⁵³. Similarly, the triazolate groups in these MOFs can potentially act as π -acceptors to enhance the electronic polarizability of the metal centre^{54–56}. With these two types of parent MOFs in hand, we then performed post-synthetic modifications to install metal–sulfur linkages.

In M_2Cl_2BBTA ($M = Co, Ni$), chloride serves as a bridging ligand between adjacent metal centres, providing a well-defined, substitutionally labile bridge that can be targeted easily to access other functionalities. During our initial attempts at replacing chloride in the presence of H₂S, we observed slow and incomplete conversion to the SH variant (for example, ~50% Cl replaced by SH after 48 h). We presume this limited reactivity is due to the formation of the by-product HCl, which is a strong acid and drives the reaction backward. To circumvent this issue, we elected to treat the chloride-based MOFs with KOH to transform the bridging chloride anions into bridging hydroxyl groups, with the MOF products denoted as $M_2(OH)_2BBTA$. Replacement of the chloride was confirmed by X-ray photoelectron spectroscopy (XPS) analysis (Supplementary Figs. 9 and 12), while the integrity of the frameworks was confirmed by powder X-ray diffraction (PXRD) analysis (Supplementary Figs. 1 and 2) and N₂ sorption isotherms collected at 77 K (Supplementary Fig. 6). We reasoned that bridging hydroxyl groups would make the subsequent sulfidation reaction more feasible as water would be the leaving group. Indeed, reacting $M_2(OH)_2BBTA$ with in situ-generated H₂S converted the lightly coloured MOF powders to dark brown within minutes, with ultraviolet–visible (UV–vis) spectra showing broad absorption bands in the visible-light range (Supplementary Fig. S6) and the ν_{S-H} bands appears on the infrared spectra (Supplementary Fig. S7), signalling the formation of the SH-bridging analogues $M_2(SH)_2BBTA$ (Fig. 2). XPS (Supplementary Tables 6 and 9) and energy-dispersive X-ray spectroscopic (EDS) analysis (Supplementary Figs. 21 and 23) revealed the anticipated 1:1 ratio between metal and sulfur, which suggests complete replacement of the hydroxyls with sulfhydryls. Importantly, scanning electron microscopy (SEM) images confirmed that these post-synthetic modifications did not affect the particle size or morphology (Supplementary Figs. 20 and 22). We note that the Co:S ratio can exceed 1:1 when excess H₂S is used (Supplementary Fig. 21), which can be attributed to the formation of polysulfides in the MOF cavities.

For the M_2X_2BBTA series ($M = Co, Ni$; $X = Cl, OH, SH$), single-crystal X-ray diffraction (SCXRD) or electron diffraction experiments enabled us to determine the solid-state structures of these MOFs. All structures were solved under the $R\bar{3}m$ space group with similar unit cell parameters (Supplementary Table 1), indicating that the parent MOFs' topology was preserved through these transformations. The simulated PXRD patterns closely match those of the experimental data (Fig. 3a and Supplementary Figs. 1 and 2), which confirms the phase purity of the bulk samples. Examination of the coordination sphere revealed that all metal ions are octahedrally coordinated and surrounded by three nitrogen atoms from different triazolate linkers. Two additional sites are occupied by X atoms from bridging groups, and the final site by terminal water molecules, except in the SH version. Notably, in the SH version, we discovered a disordered S₂ chain bridging two cobalt atoms across two distinct 1D metal chains. This corresponds well with the high Co:S ratios observed when excess H₂S was used in the synthesis (Supplementary Fig. 21). The sulfur contents of the sulfided samples were also confirmed by inductively coupled plasma–optical emission spectroscopy (ICP-OES) (Supplementary Table 18). The Co–N bond distances shortened from approximately 2.1 Å to 2.0 Å and 1.9 Å as the bridging groups changed from –Cl to –OH to –SH, respectively. This indicates that varying the bridging group alters the interactions

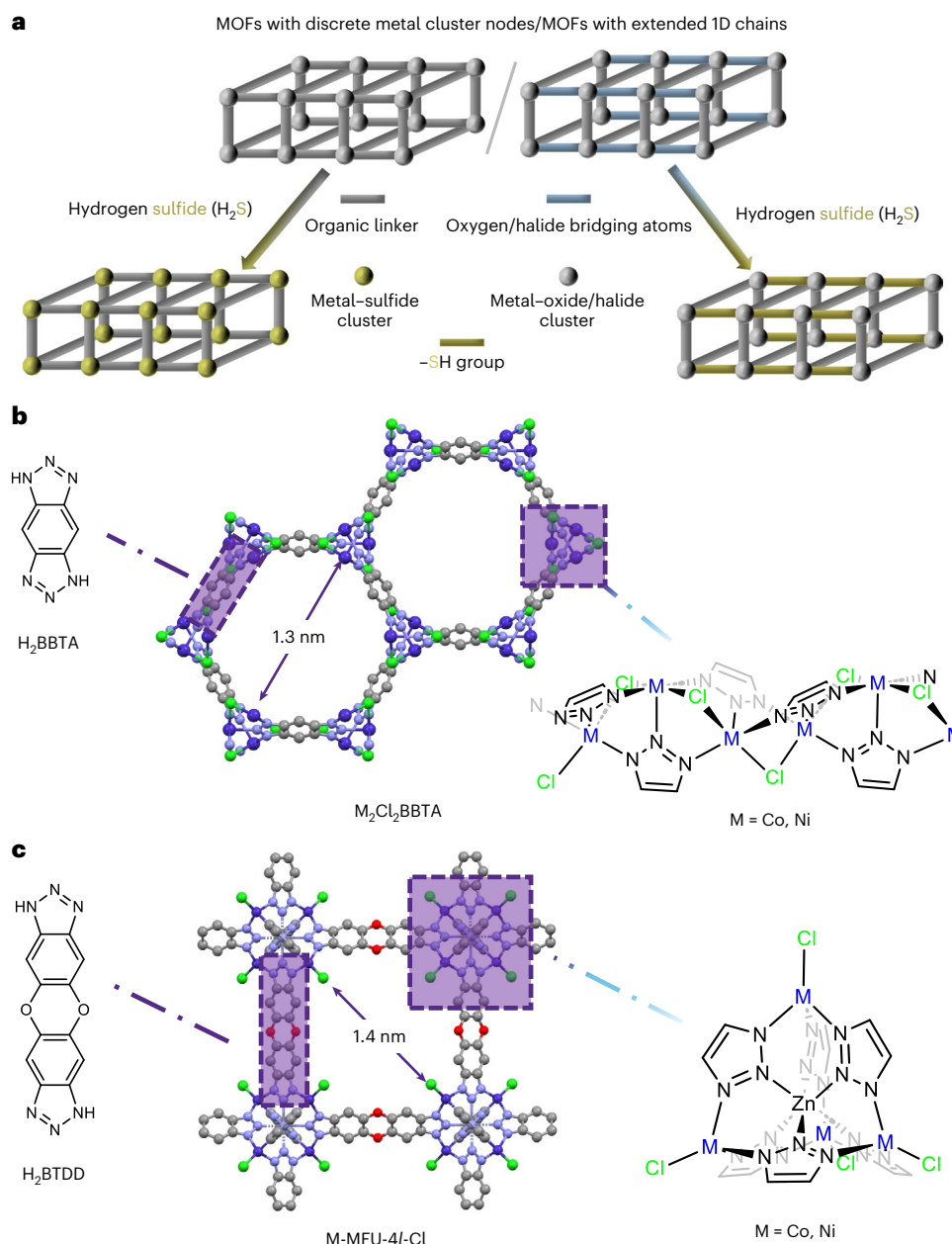


Fig. 1 | Strategy and MOF platforms used for sulfur incorporation.

a, Installation of polarizable sulfur moieties to metal-organic frameworks via post-synthetic modification by exchanging metal bridges or incorporating sulfide into discrete clusters. **b,c**, Linker, channel and metal node structures of

$\text{M}_2\text{Cl}_2\text{BBTA}$ with 1D metal chains (**b**) and M-MFU-4l-Cl with discrete metal cluster nodes (**c**). Colour code: grey, carbon; red, oxygen; yellow, sulfur; blue, nitrogen; pale blue, metal.

between the metal centres and the coordinating triazole moiety, affecting the metal-ligand bonding environment. In contrast, the Ni-N bond distances remained relatively consistent, around 2.1 Å across all versions, suggesting less variation in the metal-triazole interactions for the nickel-based structures. Next, we collected N_2 adsorption isotherms of six samples ($\text{M}_2\text{X}_2\text{BBTA}$, where $\text{M} = \text{Co}, \text{Ni}$ and $\text{X} = \text{Cl}, \text{OH}, \text{SH}$) at 77 K to analyse the intrinsic porosity of these materials. The total N_2 uptakes of $\text{M}_2\text{Cl}_2\text{BBTA}$ are STP 312 and $330 \text{ cm}^3 \text{ g}^{-1}$ with calculated Brunauer-Emmett-Teller (BET) areas of $1,180 \text{ m}^2 \text{ g}^{-1}$ and $1,190 \text{ m}^2 \text{ g}^{-1}$ (Fig. 3c) for $\text{M} = \text{Co}$ and Ni , respectively, which corresponds well with the reported values and the theoretical values calculated from crystal structures (Supplementary Table 3). The total N_2 uptakes of $\text{M}_2(\text{SH})_2\text{BBTA}$ are STP 200 and $228 \text{ cm}^3 \text{ g}^{-1}$ with calculated BET surface areas of 650 and $730 \text{ m}^2 \text{ g}^{-1}$ for $\text{M} = \text{Co}$ and Ni , respectively,

which correspond well with the theoretical values calculated from crystal structures with one terminal water occupying the open metal site. Combined, these results demonstrate that the intrinsic porosity of the framework is preserved after this post-synthetic modification protocol.

Similarly, the conversion of M-MFU-4l-Cl to M-MFU-4l-SH (where $\text{M} = \text{Co}, \text{Ni}$) utilized the same stepwise methodology as in the BBTA version (that is, conversion of Cl to OH to SH ; Supplementary Information, section B). First, the Co-MFU-4l-Cl or Ni-MFU-4l-Cl were synthesized upon incubation of the Zn-MFU-4l-Cl in dimethylformamide (DMF) solutions of the corresponding metal-chloride salt according to a previously published procedure^{50–52,57}. The chemical formulas for these MOFs were determined to be $\text{Co}_4\text{ZnCl}_4(\text{BTDD})_3$ and $\text{Ni}_{2.5}\text{Zn}_{2.5}\text{Cl}_4(\text{BTDD})_3$ through ICP-OES and XPS analyses, which

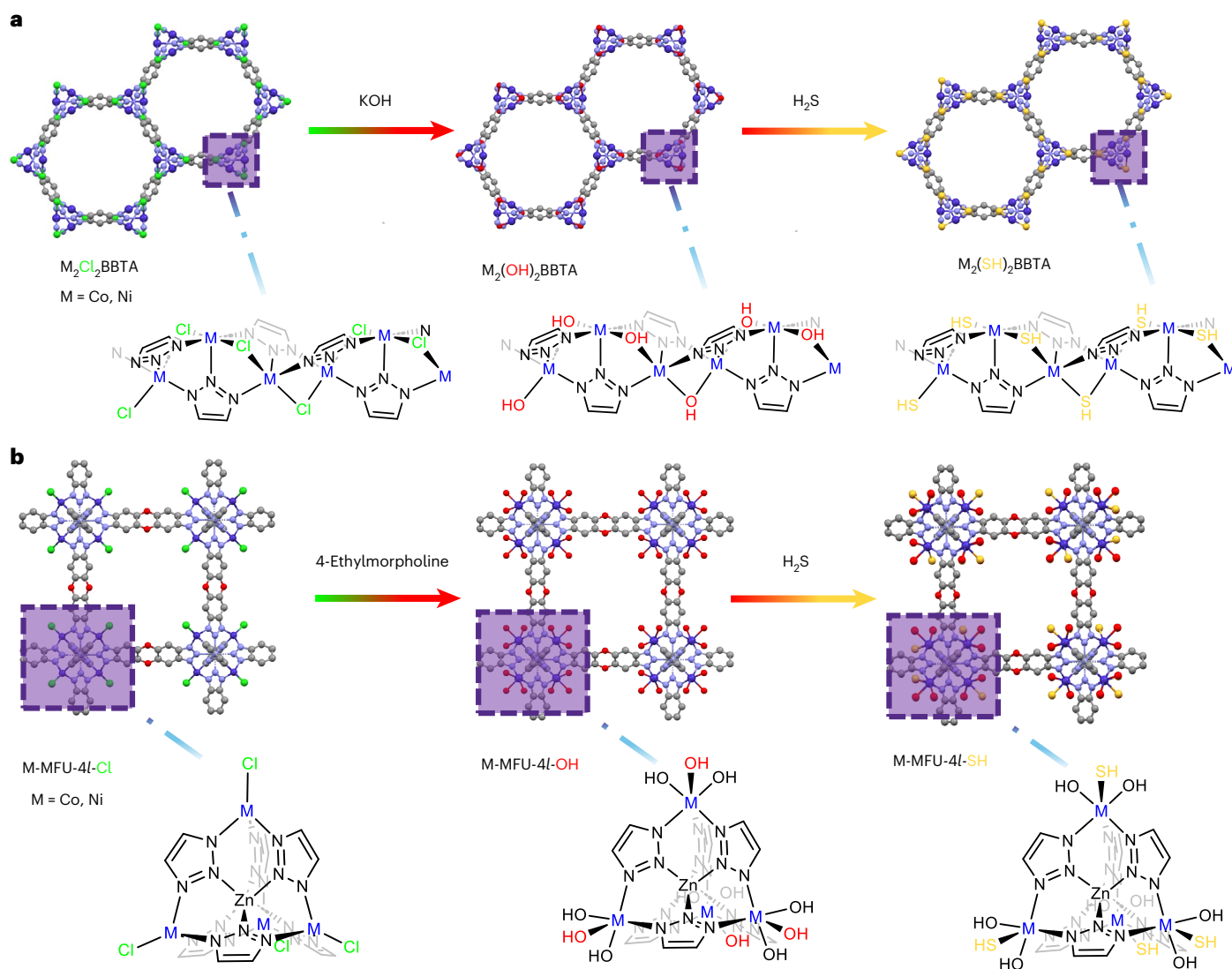


Fig. 2 | Stepwise post-synthetic routes to embed metal–sulfide active sites into MOFs. a,b, Synthesis of $M_2(SH)_2BBTA$ (**a**) and $M-MFU-4l-SH$ (**b**) ($M = Co, Ni$) via post-synthetic modifications. In **b**, the disorder arising from the coexistence of tetrahedral and octahedral coordination spheres in the $M-MFU-4l$ /metal nodes is

depicted with only one type for clarity. Hydrogen atoms are omitted for clarity. Colour code: grey, carbon; red, oxygen; yellow, sulfur; blue, nitrogen; pale blue, metal.

corresponds well with the results in the literature³². Next, we converted chlorides to hydroxyl groups through the use of a 1 M aqueous solution of organic base 4-ethylmorpholine instead of KOH because the former prevented the degradation of the cobalt- and nickel-based MOFs. Finally, treatment of these hydroxyl-based MOFs with in situ-generated H_2S afforded the anticipated colour change of the crystals to dark brown (Supplementary Fig. S6), indicating transformation of the hydroxyls to thiols. SCXRD analysis revealed that the framework structure remained largely intact post-conversion to the thiol analogues with only minor changes observed in the unit cell parameters (Supplementary Table 1). The M sites of $M-MFU-4l-SH$ are in an octahedral coordination sphere. Three coordination bonds are occupied by nitrogen atoms from the triazolyl groups of the linkers, two by oxygen atoms from coordinating water molecules, and one by a sulfur atom from the $-SH$ group (Fig. 2b). Positional disorders of coordinating oxygen and sulfur atoms are observed due to the three-fold axis imposed by the crystallographic symmetry. Additionally, the experimental PXRD patterns (Fig. 3b and Supplementary Figs. 3 and 4) closely match those that were simulated from the crystal structures, confirming the phase purity of the bulk samples. The N_2 adsorption isotherms of

these six samples at 77 K are shown in Fig. 3d and confirm the permanent porosity of these samples. The theoretical values of N_2 uptake calculated from single-crystal structures match well with the experimental values (Supplementary Table 3) for $M-MFU-4l-X$ where $M = Co, Ni$ and $X = Cl, OH$ and SH , which confirm retention of the intrinsic porosity of the initial frameworks.

In situ PDF analysis of sulfidation process

Crystallographic analysis for both variants ($M = Co, Ni$) showed that the post-synthetic functionalization results in the same topology and space group, which suggests that exchanging the bridging ligands is associated with a local rearrangement that leads to the overall same long-range order. To confirm this retention of long-range order, we further studied Co_2X_2BBTA samples using in situ PDF measurements. Specifically, these unique in situ total scattering experiments followed the local rearrangement during the dechlorination of Co_2Cl_2BBTA to $Co_2(OH)_2BBTA$ and the subsequent sulfidation of $Co_2(OH)_2BBTA$ to $Co_2(SH)_2BBTA$.

Initially, a sample of Co_2Cl_2BBTA was packed in a glass capillary and assembled in a liquid flow reactor under constant flow of 1 M KOH

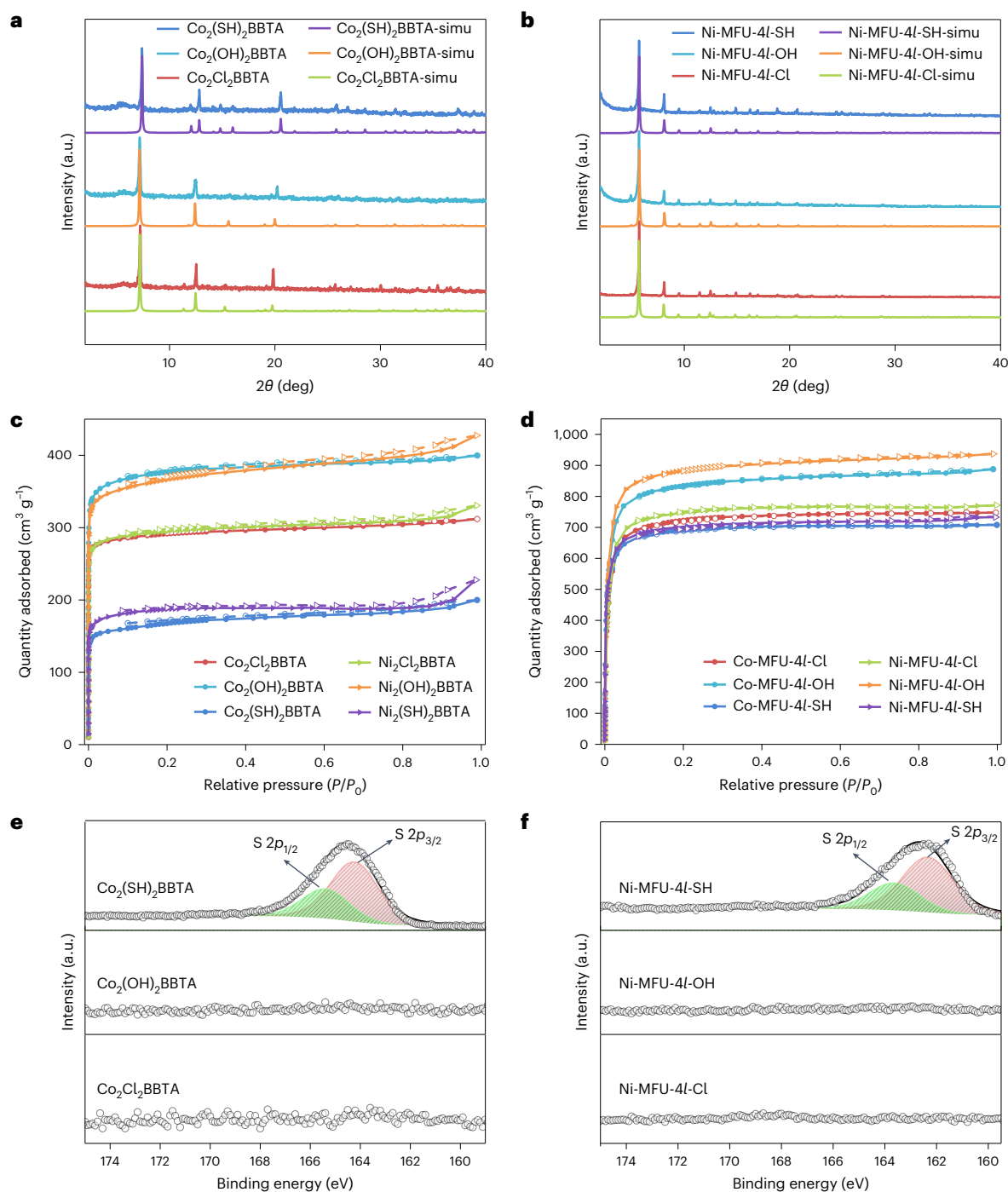


Fig. 3 | Structural integrity and sulfur incorporation characterizations across MOF series. a, b, Experimental and simulated PXRD patterns of $\text{Co}_2\text{X}_2\text{BBTA}$ (a) and Ni-MFU-4l-X (b) MOFs. **c, d,** N_2 isotherms at 77 K of $\text{M}_2\text{X}_2\text{BBTA}$ (c) and M-MFU-4l-X (d). **e, f,** XPS spectra of the S 2p peaks of $\text{Co}_2\text{X}_2\text{BBTA}$ (e) and Ni-MFU-4l-X (f) (M = Co, Ni; X = Cl, OH, SH).

to initiate the dechlorination⁵⁸. Following this process using fast PDF data acquisition enabled the direct observation of time-dependent structural changes in the local order of the framework material. We observed that local changes occur almost immediately upon contact with the highly concentrated KOH solution (for example, within ~30 s; Fig. 4a). The delay of about 15 s after the start of the experiment is assigned to dead volume in the sample environment upstream of the sample. The rapid conversion of $\text{Co}_2\text{Cl}_2\text{BBTA}$ to $\text{Co}_2(\text{OH})_2\text{BBTA}$ was also confirmed by ex situ PDF studies using a more dilute media, which allowed us to track the rearrangement at a higher time resolution. After five exposure cycles to minimal amounts of

KOH (0.1 μl of 0.01 M), ligand exchange at the local order of the metal nodes is observed from changes in peak intensities in the PDF data in which the Co–Cl peak decreases in intensity and Co–O peak rises (Fig. 4b).

The sulfidation of $\text{Co}_2(\text{OH})_2\text{BBTA}$ was initiated through in situ generation of H_2S using NaSH and acidified water as precursor, which was directed over the sample using nitrogen as the carrier gas. Similar to the transformation from Co–Cl to Co–OH species, the local rearrangement of the metal centres also occurred rapidly, within minutes after the gas flow reached the sample (Fig. 4c). Conversion can be verified by a shift in the peak position around 2 Å (for example, shift

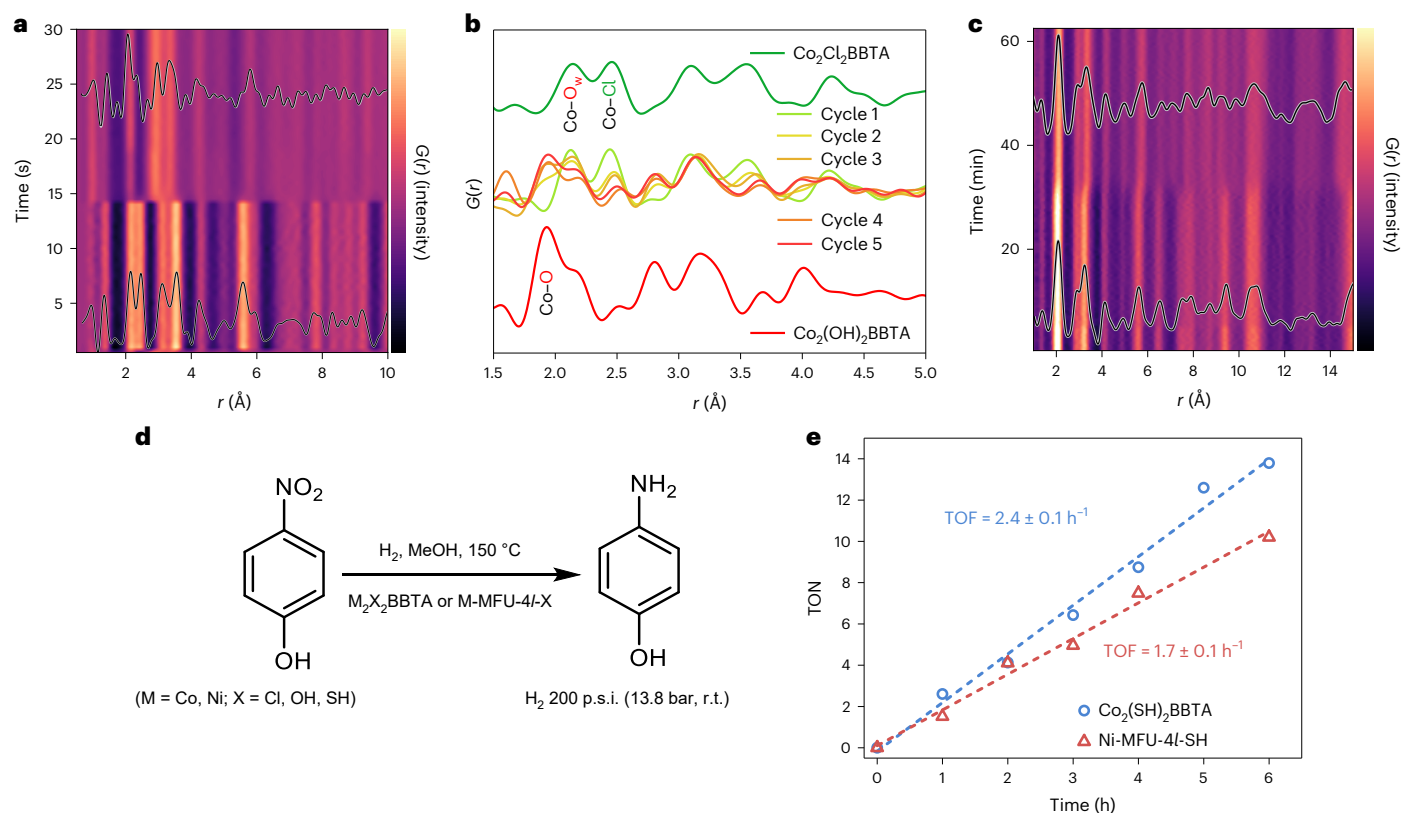


Fig. 4 | In situ structural evolution during sulfidation and catalytic hydrogenation performance. **a**, Time-dependent PDF data of $\text{Co}_2\text{Cl}_2\text{BBTA}$ under KOH flow. Each horizontal trace represents a PDF collected at a different time point (from bottom to top), with a time interval of 10 seconds. The PDF patterns of initial and final phase matching $\text{Co}_2\text{Cl}_2\text{BBTA}$ and $\text{Co}_2(\text{OH})_2\text{BBTA}$, respectively, are overlaid. **b**, PDF data of $\text{Co}_2\text{Cl}_2\text{BBTA}$ before and after cycling with 0.1 M KOH. The $\text{Co}_2(\text{OH})_2\text{BBTA}$ PDF is used as a reference; the Co–O, Co–O_w (water) and Co–Cl

bond distances are labelled. **c**, In situ PDF data of $\text{Co}_2(\text{OH})_2\text{BBTA}$ under H_2S flow. Traces are shown from bottom to top, with each representing a time point spaced 30 seconds apart. The PDF graphs of initial and final phase matching $\text{Co}_2(\text{OH})_2\text{BBTA}$ and $\text{Co}_2(\text{SH})_2\text{BBTA}$, respectively, are overlaid. **d**, Model reaction of hydrogenation of 4-nitrophenol to 4-aminophenol with molecular hydrogen. **e**, Time-dependent turnover number (TON) of $\text{Co}_2(\text{SH})_2\text{BBTA}$ (1.5 mol%) and Ni-MFU-4l-SH (5 mol%) at 150°C . Dotted lines represent the corresponding linear regressions.

from 2.05 Å to 2.13 Å) due to the difference in Co–O and Co–S bond lengths. In addition to structural verification from PDF analysis, the crystallinity and lattice symmetry were confirmed via PXRD analysis after each conversion step (Supplementary Fig. 5).

XPS analysis of sulfur incorporation

Following the successful conversion of MOFs to their sulfided versions, confirmed by X-ray techniques, XPS spectroscopy was used to further identify the oxidation state of the sulfided MOFs in the post-synthetically modified samples. For the $\text{M}_2\text{X}_2\text{BBTA}$ series, the results indicate that after the treatments with KOH, there is no chloride left in the samples (Supplementary Figs. 8, 9, 11 and 12 and Supplementary Tables 4, 5, 7 and 8). After the H_2S treatment, a peak of binding energy (BE) at -163 eV (162.7 and 162.6 eV for $\text{Co}_2(\text{SH})_2\text{BBTA}$ and $\text{Ni}_2(\text{SH})_2\text{BBTA}$, respectively) starts appearing on the XPS spectra (Fig. 3e, Supplementary Figs. 10 and 13 and Supplementary Tables 6 and 9), which falls into the range of $-\text{SH}$ functional groups with sulfur in the -2 oxidation state. However, the peak corresponding to the $2p_{3/2}$ band of cobalt only shifts in BE from 780.5 eV to 781.0 eV to 779.0 eV from $\text{X} = \text{Cl}$ to OH to SH due to the different coordination environments. The preservation of the satellite feature indicates that the oxidation state of cobalt remains $+2$ (refs. 59–61). The BE of the $2p_{3/2}$ band of $\text{Co}_2(\text{SH})_2\text{BBTA}$ is similar to those of previously reported cobalt sulfide compounds⁶². For the $\text{M} = \text{Ni}$ case, the $2p_{3/2}$ peaks of nickel shift from 855.7 eV to 854.6 eV to 853.0 eV for $\text{X} = \text{Cl}$, OH and SH , respectively. The BE of the nickel $2p_{3/2}$ band for $\text{Ni}_2(\text{SH})_2\text{BBTA}$ is similar to those of reported nickel sulfide compounds⁶². Furthermore, the XPS results for

Table 1 | Conversion of 4-nitrophenol to 4-aminophenol via hydrogenation using molecular hydrogen and different MOFs

MOFs with 1D metal chains			
$\text{Co}_2\text{X}_2\text{BBTA}$	% conversion	$\text{Ni}_2\text{X}_2\text{BBTA}$	% conversion
$\text{X} = \text{Cl}$	2	$\text{X} = \text{Cl}$	8
$\text{X} = \text{OH}$	9	$\text{X} = \text{OH}$	18
$\text{X} = \text{SH}$	>99	$\text{X} = \text{SH}$	37
MOFs with discrete 3D metal nodes			
Co-MFU-4l-X	% conversion	Ni-MFU-4l-X	% conversion
$\text{X} = \text{Cl}$	4	$\text{X} = \text{Cl}$	8
$\text{X} = \text{OH}$	1	$\text{X} = \text{OH}$	20
$\text{X} = \text{SH}$	23	$\text{X} = \text{SH}$	>99

the M-MFU-4l-X series showed similar patterns (Fig. 3f). For instance, the treatment of M-MFU-4l-Cl (M = Co, Ni) with the organic base to form M-MFU-4l-OH efficiently removed the chloride in the samples (Supplementary Figs. 14, 15, 17 and 18 and Supplementary Tables 10, 11, 13 and 14). Similarly, following treatment of M-MFU-4l-OH with H_2S , the peaks that appear in the XPS spectra around 162 eV demonstrate the successful incorporation of $-\text{SH}$ group into the framework and the formation of M-MFU-4l-SH (Fig. 3f, Supplementary Figs. 16 and 19 and Supplementary Tables 12 and 15).

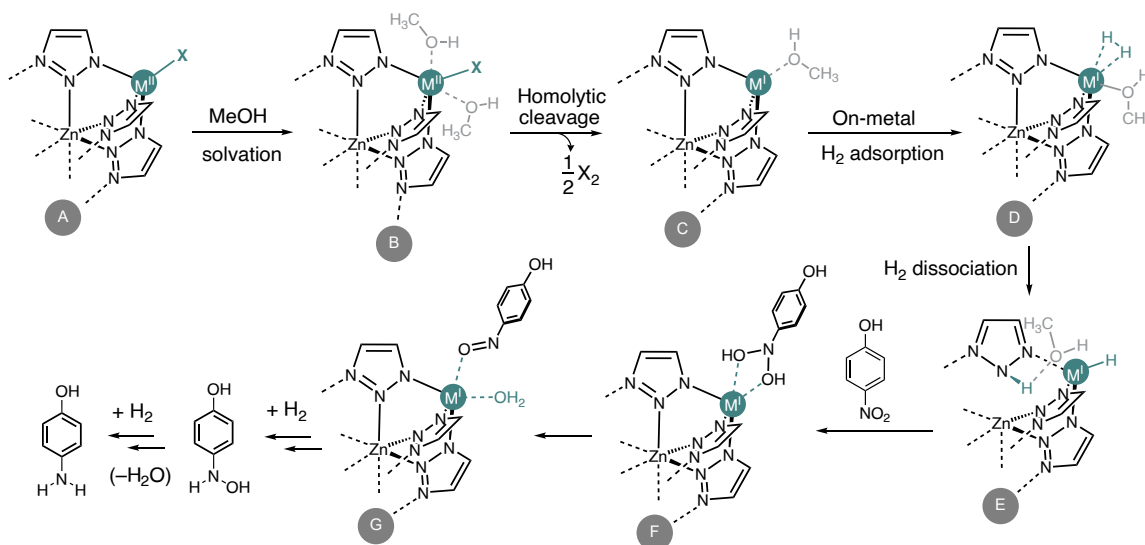
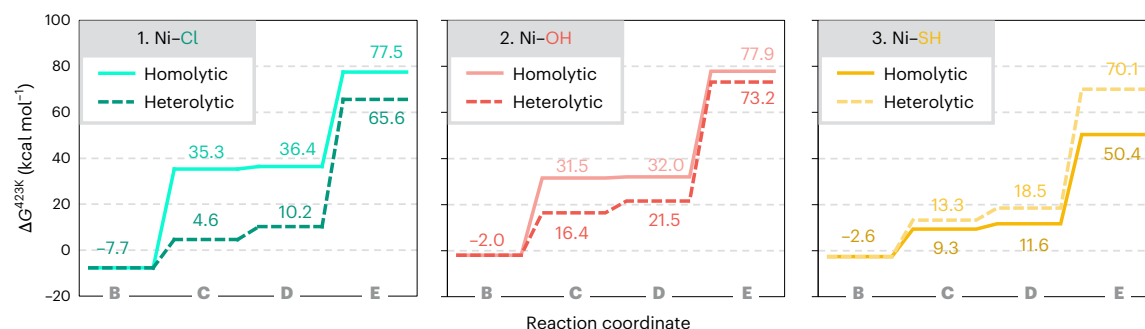
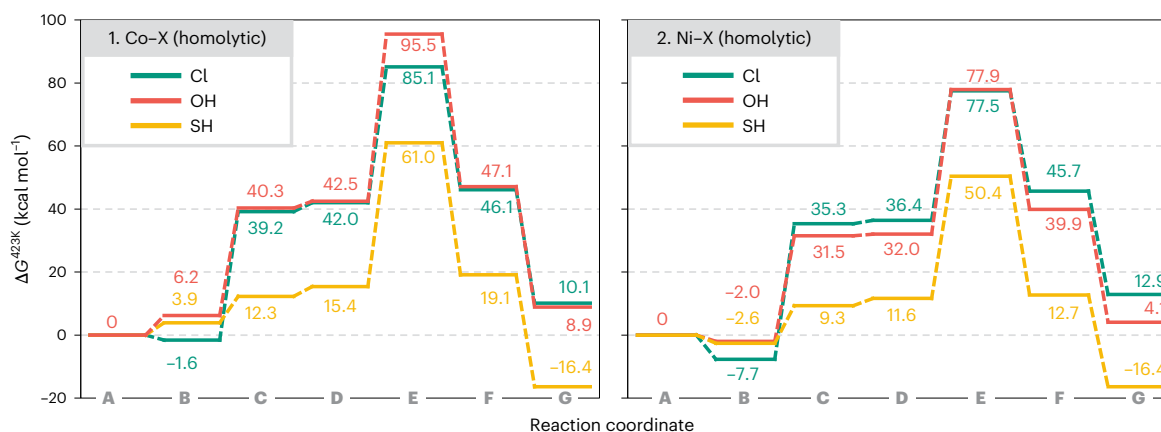
a Proposed mechanism**b** Energetics for active species **E** formation (MFU-4l)**c** Energetics for the reduction of Ar-NO₂ to Ar-NO (MFU-4l)

Fig. 5 | Computed mechanisms and energetics of hydrogenation catalysis in MOF-based systems. **a**, Schematic of open metal-site generation, hydrogen adsorption and dissociation for subsequent Ar-NO₂ reduction in the M-MFU-4l catalyst (M = Co, Ni). The seven corresponding species are denoted as **A–G**. **b**, Free-energy profiles (referenced to unsolvated species **A**) for active species **E** formation in Ni-MFU-4l with different Ni-X bonds (X = Cl, OH, SH), comparing

homolytic and heterolytic cleavage pathways. **c**, Free-energy profiles for Ar-NO₂ reduction to Ar-NO catalysed by MFU-4l with (1) Co-X and (2) Ni-X active sites (X = Cl, OH, SH), highlighting the energetic favourability of the SH ligand in the homolytic pathway. In **b** and **c**, Gibbs free energies are reported in kcal mol⁻¹ and were calculated at the M06-D3(0)/basis-II/SMD(MeOH)//TPSSH-D3(BJ)/basis-II level of theory.

Hydrogenation catalysis with sulfided MOFs

To test the catalytic performance of the sulfur-incorporated MOFs towards hydrogenation reactivity, the reduction of 4-nitrophenol to 4-aminophenol was selected as a model reaction. All 12 MOFs (M₂X₂BBTA and M-MFU-4l-X, where M = Co, Ni and X = Cl, OH, SH) were tested under the same conditions at 150 °C for 4.25 h under 13.8 bar (r.t.) H₂ (Fig. 4d)

in a Parr vessel. The results of the reactions are summarized in Table 1. The M₂Cl₂BBTA MOFs showed minimal conversion (that is, close to that of the background reaction without the inclusion of MOFs; Supplementary Figs. 29 and 32), and the conversion rate improved slightly for both M₂(OH)₂BBTA samples with ~9% and 18% conversion for M = Co and Ni, respectively. In contrast, M₂(SH)₂BBTA exhibited a notable improvement

in the conversion rate to ~37% for nickel and ~100% for cobalt. Interestingly, for the M-MFU-4l-X series, M = Co showed relatively low conversion when X = Cl (~4%), X = OH (~1%) and even when X = SH (~23%). In contrast, the Ni-MFU-4l-X analogues demonstrated relatively higher conversion rates (~8% for X = Cl and ~20% for X = OH), and Ni-MFU-4l-SH achieved >99% conversion under these conditions. Cycling studies revealed that both Co₂(SH)₂BBTA and Ni-MFU-4l-SH showed consistent conversion rates of >99% under the same conditions. Next, we studied the kinetics of these reactions with lower catalyst loadings (Fig. 4e). At 1.5 mol% loading, Co₂(SH)₂BBTA achieved a turnover frequency (TOF) of $2.4 \pm 0.1 \text{ h}^{-1}$. However, the TOF for Ni-MFU-4l-SH at the 5% catalyst loading was measured to be $1.7 \pm 0.1 \text{ h}^{-1}$ under analogous conditions.

Mechanistic elucidation with density functional theory calculations

To elucidate the experimentally observed reactivity trends in nitroarene hydrogenation, we performed density functional theory (DFT) calculations. Starting with extended crystals constrained by periodic boundary conditions, we constructed finite-sized cluster models by truncating the framework around a metal node (Extended Data Fig. 1). Using these cluster models, we propose a reaction pathway that begins with the coordination of methanol solvent molecule(s) to the metal centres, followed by M–X bond cleavage to generate an open metal site, H₂ adsorption and subsequent H–H bond dissociation, ultimately forming the active species **E** (Fig. 5a) responsible for reduction catalysis^{63,64}.

Experimentally, we found that methanol is critical for the catalysis. To probe its role, we systematically explored methanol coordination effects (Supplementary Fig. 42) and observed distinct solvation behaviours across different MOFs. For example, the Co(II) centre in MFU-4l is predicted to remain tetrahedral (unsolvated), whereas the Ni(II) centre should adopt an octahedral geometry (solvated). Moreover, we propose a methanol-assisted pathway for H₂ dissociation and the formation of active species **E** (Fig. 5a, Extended Data Fig. 2 and Supplementary Information, sections H.5–H.5) consistent with the hydrogenation catalysis observed in cobalt single-atom systems⁶⁴.

The formation of open metal sites was investigated by exploring both homolytic (C1) and heterolytic (C2) M–X bond cleavage routes (Fig. 5a and Extended Data Fig. 2). Computational results show that the SH ligand facilitates homolytic cleavage, with the MFU-4l framework demonstrating a preference for the homolytic route over the heterolytic one ($\Delta G_{\text{C1-C2}} = -4.0 \text{ kcal mol}^{-1}$; Fig. 5b). In contrast, Cl and OH ligands favour the heterolytic pathway, with $\Delta G_{\text{C1-C2}}$ values of 30.7 and 15.1 kcal mol⁻¹, respectively.

Following open metal site generation, H₂ adsorption and subsequent methanol-mediated dissociation leads to the formation of active species **E**. For the Ni-MFU-4l-X series, the SH group enables the most facile generation of **E** ($\Delta G^{423\text{K}}(\text{E}) = 53.0 \text{ kcal mol}^{-1}$; Fig. 5b), whereas Cl and OH groups require higher energy with $\Delta G^{423\text{K}}(\text{E})$ values of 73.3 and 75.2 kcal mol⁻¹, respectively. This reactivity trend across X groups persists for Co-MFU-4l systems, with SH remaining the most reactive ligand (Fig. 5c(1)). However, Co-MFU-4l-SH ($\Delta G^{423\text{K}}(\text{E}) = 61.0 \text{ kcal mol}^{-1}$) is predicted to exhibit lower reactivity than its nickel analogue ($\Delta G^{423\text{K}}(\text{E}) = 53.0 \text{ kcal mol}^{-1}$), consistent with experimental observations.

A similar mechanism was invoked in the 1D BBTA system (Extended Data Fig. 2), which also shows preference for the SH group over Cl or OH for both cobalt and nickel variants (Supplementary Fig. 54). Interestingly, within the BBTA framework, our calculations predict that the Co–SH catalyst generates species **E** more favourably than the Ni–SH analogue, with $\Delta G^{423\text{K}}(\text{E})$ values of 62.4 and 67.6 kcal mol⁻¹, respectively. These results again corroborate the experimentally observed higher conversion rates for Co–SH compared with Ni–SH in the 1D BBTA framework.

To better understand the reactivity trends of various X ligands (X = Cl, OH, SH) bound to metal centres M (= Co, Ni), we performed energy decomposition analysis (EDA) based on absolutely localized molecular orbitals (ALMOs)^{65,66}. This approach enabled a detailed analysis of the contributions to M^{II}–X bonding in MFU-4l–Ni–X systems, as detailed in Supplementary Information, section H.10 and Extended Data Table 1. The total electronic contribution (ΔE_{tot}) for binding X to a formal M(I) site showed increasing M–X interaction strength from SH to OH to Cl, with binding energies (in kcal mol⁻¹) of –72.0 (SH), –83.2 (OH) and –94.4 (Cl). This trend was dominated by the charge-transfer term (ΔE_{CT}), which quantifies energy lowering from orbital mixing between X and M(I) fragments, showing increasingly favourable ΔE_{CT} values (in kcal mol⁻¹) of –59.9 (SH), –85.9 (OH) and –92.7 (Cl). These results indicate reduced electron density sharing between the metal centre and SH ligand compared with OH or Cl, resulting in a more labile M^{II}–SH bond with enhanced propensity for generating open metal sites. Consequently, the ease of homolytic M^{II}–X bond cleavage and open metal site generation decreases in the order: SH > OH > Cl.

We further explored the differential reactivity of Co^{II}–SH and Ni^{II}–SH active sites in BBTA and MFU-4l frameworks by analysing bonding interactions between open metal sites (M^I; M = Co, Ni) and H₂ molecules. Using ALMO-EDA analysis, we found that coordination geometry plays a decisive role in determining catalytic performance (Supplementary Information, section H.10 and Extended Data Table 2). In the pseudo-octahedral environment of BBTA, Co(I) (*d*⁸) with two singly occupied *e_g* orbitals experiences lower Pauli repulsion (13.5 kcal mol⁻¹) than Ni(I) (*d*⁹) with one doubly and one singly occupied *e_g* orbital (51.8 kcal mol⁻¹), resulting in stronger H₂ adsorption at Co(I) centres (–5.8 kcal mol⁻¹ versus –3.7 kcal mol⁻¹ for Ni(I)). Conversely, in the pseudo-tetrahedral MFU-4l framework, this trend reverses due to inverted crystal field splitting, with Ni(I) showing lower Pauli repulsion (32.0 kcal mol⁻¹ versus 39.4 kcal/mol) and stronger H₂ binding (–20.9 kcal mol⁻¹ versus –14.9 kcal mol⁻¹) than Co(I). These findings rationalize the observed reactivity patterns: cobalt performs better in octahedral BBTA environments, whereas nickel demonstrates superior activity in tetrahedral MFU-4l frameworks.

Conclusion

We developed a stepwise post-synthetic strategy for synthesizing MOFs with highly polarizable SH groups by converting M–Cl species to M–OH and subsequently to M–SH (M = Co, Ni), overcoming challenges in the direct synthesis of metal–sulfur MOFs. This method was successfully applied to both 1D M₂(SH)₂BBTA and 3D M-MFU-4l-SH series, with SCXRD and PDF analyses confirming quantitative conversion. Catalytic studies on 4-nitrophenol hydrogenation demonstrated that incorporating sulfur-based ligands substantially enhances reactivity compared to chloride- and hydroxide-based analogues. DFT calculations further showed that the lability of the metal–sulfur bond plays a crucial role in the M–X bond cleavage pathway, enhancing hydrogen activation and offering valuable insights for future MOF-based catalyst design. This approach establishes a versatile method for integrating polarizable sulfur components into porous materials, broadening opportunities for high-performance catalyst development.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41557-025-01876-y>.

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Methods

Materials and general methods

All reagents were obtained from commercial suppliers and were utilized as received without additional purification.

The powder samples of $\text{Co}_2\text{Cl}_2\text{BBTA}$ and $\text{Ni}_2\text{Cl}_2\text{BBTA}$ were synthesized by methods reported in the literature⁴⁹.

Synthesis of the powder samples of $\text{M}_2\text{Cl}_2\text{BBTA}$

First, 7 ml of a DMF solution of H_2BBTA (1*H*,5*H*-benzo(1,2-*d*:4,5-*d'*)bistriazole, 30 mg, 0.19 mmol) and 7 ml of a methanol solution of $\text{MCl}_2 \cdot 6\text{H}_2\text{O}$ ($\text{M} = \text{Co}, \text{Ni}$; 100 mg, 0.42 mmol) were combined in an 8-dram vial. Concentrated HCl (12 M) was then added (0.1 ml for cobalt and 3 ml for nickel). The mixture was incubated at 65 °C for 72 h. The resulting powder was washed three times with DMF and three times with methanol. To remove any residual DMF within the cavity, Soxhlet extraction with methanol was conducted twice, each for 36 h. Activation was conducted under dynamic vacuum at 120 °C for 24 h before adsorption measurements.

Synthesis of the powder samples of $\text{M}_2(\text{OH})_2\text{BBTA}$

$\text{M}_2\text{Cl}_2\text{BBTA}$ (100 mg) is dispersed in 100 ml KOH (1 M) aqueous solution under nitrogen for 12 h. The product was washed with water and methanol three times and dried under a nitrogen flow. Activation was conducted under dynamic vacuum at 120 °C for 24 h before adsorption measurements. An alternative method⁶⁷ for the preparation of $\text{Co}_2(\text{OH})_2\text{BBTA}$ was also tested using NH_3 as a base to assess the potential formation of metal hydroxides in this system. No substantial differences were observed in subsequent experiments or catalytic results.

Synthesis of the powder samples of $\text{Co}_2(\text{SH})_2\text{BBTA}$

$\text{Co}_2(\text{OH})_2\text{BBTA}$ (30 mg) was dispersed in 4 ml of methanol in a 4-dram vial containing a stir bar. The mixture was sealed and exposed to in situ-generated H_2S , produced from 19 mg of NaHS and 0.5 ml of concentrated H_3PO_4 , for 5 h. After exposure, the product was washed three times with methanol and dried under a flow of nitrogen. The sample was then activated at 100 °C for 24 h prior to conducting adsorption measurements.

Synthesis of the powder samples of $\text{Ni}_2(\text{SH})_2\text{BBTA}$

$\text{Ni}_2(\text{OH})_2\text{BBTA}$ (30 mg) was dispersed in 4 ml of methanol in a 4-dram vial containing a stir bar. The mixture was sealed and exposed to in situ-generated H_2S , produced from 19 mg of NaHS and 0.5 ml of concentrated H_3PO_4 , for 12 h. After exposure, the product was washed three times with methanol and dried under a flow of nitrogen. The sample was then activated at 100 °C for 24 h prior to conducting adsorption measurements.

Synthesis of the single-crystal samples of $\text{Co}_2\text{Cl}_2\text{BBTA}$

A synthesis mixture was prepared by combining 4 ml of a solution of H_2BBTA (20 mg, 0.125 mmol) in diethylformamide, 4 ml of a methanol solution of $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (60 mg, 0.252 mmol) and 0.2 ml of 38% HCl in a 6-dram vial. This vial was then placed in a preheated oven at 70 °C. After 3 days, needle-like pink single crystals had formed.

Post-synthetic conversion of the single-crystal samples of $\text{Co}_2\text{Cl}_2\text{BBTA}$ to $\text{Co}_2(\text{OH})_2\text{BBTA}$

A single-crystal sample of $\text{Co}_2\text{Cl}_2\text{BBTA}$ (~5 mg) was treated with 4 ml KOH (0.1 M) for 4 h at room temperature and then washed with deionized water three times. The morphology of the crystals remains the same but their colour becomes darker.

Post-synthetic conversion of the single-crystal samples of $\text{Co}_2(\text{OH})_2\text{BBTA}$ to $\text{Co}_2(\text{SH})_2\text{BBTA}$

A single-crystal sample of $\text{Co}_2(\text{OH})_2\text{BBTA}$ (~3 mg) was dispersed in 2 ml methanol and treated with in situ-generated H_2S (generated by mixing

20 mg NaHS with 0.5 ml concentrated H_3PO_4) for 4 h at room temperature and then washed with methanol 3 times. The morphology of the crystals remains the same but their colour becomes darker.

Synthesis of powder samples of M-MFU-4l-SH

M-MFU-4l-OH (30 mg) was dispersed in 4 ml methanol. The mixture was sealed and exposed to in situ-generated H_2S , produced from 50 mg of NaHS and 0.5 ml of concentrated H_3PO_4 , for 12 h. Subsequently, the brown sample was washed with water three times. The resulting powder sample was dried under a nitrogen flow and activated under 120 °C before adsorption measurements.

Post-synthetic conversion of the single-crystal samples of M-MFU-4l-OH to M-MFU-4l-SH

Single-crystal samples of M-MFU-4l-OH (~2 mg) were dispersed in 4 ml methanol. The mixture was sealed and exposed to in situ-generated H_2S , produced from 50 mg of NaHS and 0.5 ml of concentrated H_3PO_4 , for 12 h. The morphology of the crystals remained the same but their colour changed to dark brown.

SCXRD data of $\text{Co}_2\text{Cl}_2\text{BBTA}$, $\text{Co}_2(\text{OH})_2\text{BBTA}$, $\text{Co}_2(\text{SH})_2\text{BBTA}$, Co-MFU-4l-Cl, Co-MFU-4l-OH, Co-MFU-4l-SH, Ni-MFU-4l-Cl, Ni-MFU-4l-OH, and Ni-MFU-4l-SH were collected at 100 K, using a Rigaku Cu-Synergy/DW-Synergy diffractometer, equipped with a shutterless electronic-noise free Hybrid Photon Counting detector, a Cryostream 80-500 K cooler (Cryosystems), a Cu K α ($\lambda = 1.54184 \text{ \AA}$) microfocus source with a beam size of $\sim 110 \mu\text{m}$ and a four-circle kappa goniometer. The single crystals were mounted on MicroMesh (MiTeGen) with paratone oil. ED data for $\text{Ni}_2\text{Cl}_2\text{BBTA}$, $\text{Ni}_2(\text{OH})_2\text{BBTA}$ and $\text{Ni}_2(\text{SH})_2\text{BBTA}$ were collected at 293 K under vacuum using MicroED, also known as the XtaLAB Synergy-ED system. The instrument is equipped with a Rigaku HyPix-ED direct electron detector and controlled by the Rigaku Oxford Diffraction CrysAlisPro software package. The XtaLAB Synergy-ED system utilizes a 200-kV electron gun, featuring a lanthanum hexaboride crystal as the electron emitter for optimized electron optics. The structures were determined by intrinsic phasing (SHELXT 2018/2) and refined by full-matrix least-squares refinement (SHELXL-2018/3) using the Olex2 software packages. The disordered non-coordinated solvents were removed using the solvent marks option in Olex2 software. The refinement results are summarized in Supplementary Table 1. Crystallographic data crystal structures in CIF format have been deposited in the Cambridge Crystallographic Data Centre (CCDC) under the deposition numbers shown in Supplementary Table 1.

PXRD measurements of MOFs were conducted at room temperature using a STOE-STADIMP powder diffractometer. This set-up included an asymmetric curved germanium monochromator utilizing Cu K α 1 radiation ($\lambda = 1.54056 \text{ \AA}$) and a 1D silicon strip detector (MYTHEN2 1K from DECTRIS). The copper X-ray tube, set in a line focus configuration, was operated at 40 kV and 40 mA. Samples of activated powder were placed between two Kapton foils and analysed in transmission mode within a rotating holder. Data collection ranged from $2\theta = 1^\circ$ – 40° over 6-min intervals. Prior to data acquisition, instrument calibration was verified using a NIST silicon standard (640d). XPS measurements were performed at the Keck-II/NUANCE facility at Northwestern University using a Thermo Scientific ESCALAB 250 Xi, which uses Al K α radiation (photon energy, 1,486.6 eV) and includes an electron flood gun. Data analysis was conducted using Thermo Scientific Avantage Data System software. All spectra were calibrated relative to the C 1s peak at 284.8 eV.

N_2 adsorption isotherms were measured at 77 K using a Micromeritics ASAP 2420. The BET surface area was calculated from the N_2 sorption data within the relative pressure range of $P/P_0 = 0.05$ – 0.2 .

Elemental analysis was performed using a Varian Vista-MDX ICP-OES spectrometer equipped with a charge-coupled device detector and argon plasma operating over the 175–785 nm wavelength range.

Approximately 5 mg of each preactivated sample was digested in 2 ml of concentrated HNO_3 using a Biotage SPX microwave reactor (software version 2.3, build 6250) at 200 °C for 30 min. After digestion, the whole resulting solution was diluted to 40 ml with Millipore water prior to analysis. Hydrogenation catalysis were carried out in Parr reactors equipped with Teflon liners, which were precleaned using aqua regia. The reactors were initially purged three times with argon at 7 bar, followed by three purges with hydrogen at 13.7 bar. Post-reaction, the products and catalysts were separated using a centrifuge. The catalysts were then washed with appropriate solvents prior to reuse.

PDF analysis

In situ X-ray scattering data, suitable for XRD and PDF analysis, were collected at beamline 28-ID-1 of the National Synchrotron Light Source II at Brookhaven National Laboratory. Samples were loaded in either borosilicate (for dichlorination experiments) or polyimide capillaries (outer diameter, ~1.1 mm) and assembled within a flow-cell apparatus that either allows for a constant flow of a solution over the sample⁵⁸ or in a gas-flow furnace that allows for heating in a desired gas environment⁶⁸. X-ray scattering images were collected using high-energy X-rays ($\lambda = 0.1653 \text{ \AA}$) at sample-to-detector distances optimized for PDF and XRD analysis.

Data were collected for ex situ samples and for in situ dechlorination and sulfidation of a Co-BBTA sample. For the dechlorination experiment the sample was loaded in a borosilicate capillary and a 1 M KOH solution was directed over the sample while data were collected at the PDF distance for 10 s per scan. The dechlorination was also studied in an ex situ approach using a more dilute KOH solution (0.01 M) where the reaction was quenched at certain intervals to visualize the changes over a longer time resolution. The in situ sulfidation using H_2S as sulfiding agent was conducted using NaHS (10 mg) as a precursor, which was primed with diluted H_2SO_4 (100 ml); evolving H_2S gas was directed over the sample using nitrogen as carrier gas, while data were collected at the PDF distance.

The sample-to-detector distance and instrument geometry were optimized within GSAS-II based on the scattering collected for a CeO_2 standard⁶⁹. Images were reduced to 1D intensity data as a function of scattering angle. The scattering contribution from the sample environment was subtracted and the data converted into PDFs, $G(r)$, within xPDFsuite ($Q_{\text{max}} = 21.48 \text{ \AA}^{-1}$)⁷⁰.

Computational methods

Periodic DFT calculations. To perform DFT calculations on cluster models, the geometries of all studied MOFs were optimized using periodic boundary conditions with the PBEsol-D3(O) density functional⁷¹ in the Vienna Ab initio Simulation Package (VASP, v.6.4.0)⁷². Dispersion interactions were accounted for using Grimme's D3 method⁷³. Valence-core interactions were described by the projector-augmented wave method^{74,75}. The valence electrons included in the calculations for each atom type were: C (4), H (1), N (5), O (6), S (6), Cl (7), Co (9), Ni (10), Zn (12). A 520-eV plane-wave energy cut-off was used, and geometries were relaxed until atomic forces were below $0.01 \text{ eV \AA}^{-1}$. Brillouin zone sampling was performed using a Γ -point only $1 \times 1 \times 1$ k -mesh. These optimized structures were used as the basis for finite-sized cluster model calculations.

Cluster calculations were performed using DFT as implemented in Gaussian 16 (rev. B.01)⁷⁶. Geometry optimizations and frequency calculations were carried out at the TPSSH-D3(BJ)/basis-I level of theory^{77,78}, where basis-I denotes def2-SVP for non-metals and def2-TZVP with an Stuttgart-Dresden (SDD) effective core potential for cobalt, nickel and zinc^{79,80}. An 'ultrafine' grid was utilized for numerical integration. The natures of all stationary points were verified by calculating quasi-harmonic vibrational frequencies. Entropic terms were obtained using the rigid-rotor-harmonic-oscillator approximation, with the free-rotor approximation applied for vibrational modes below 100 cm^{-1} (ref. 81).

Single-point electronic energies were recomputed at the M06-D3(O)/basis-II level^{82,83}, where basis-II is def2-TZVP for all atoms with an SDD effective core potential for cobalt, nickel and zinc^{79,80}, using the solvation model based on density (SMD) continuum solvation model with solvent parameters for methanol. Final Gibbs free energies (G) at 423.15 K were calculated by combining this electronic energy with thermal contributions to G from TPSSH-D3(BJ)/basis-I calculations.

Thermodynamic corrections. To account for the change of standard-state concentration from 1 atm pressure in the gas phase to 1 M in the solution phase, an entropy correction term of $1.89 \text{ kcal mol}^{-1}$ was added to the absolute free energies of all computed species. For methanol solvent, the pure liquid (24.72 mol l^{-1}) was used as the standard state. The correction to the Gibbs free energy due to the change in the standard state was calculated as follows^{84,85}:

$$G^\circ = G^{\text{latm}} + RT \ln(C^\circ / C^{\text{latm}})$$

Here, G° denotes the Gibbs free energy relative to the appropriate standard state in solution phase, G^{latm} refers to the ideal gas standard Gibbs free energy at 298.15 K under a standard state of 1 atm, C° represents the concentration in the standard reference state (specifically, 24.72 M for methanol), while C^{latm} is the concentration of an ideal gas at standard pressure (1 atm), given by $C^{\text{latm}} = 1/V_m = P/RT$. The correction term for each methanol molecule in the equation above is equal to $2.7 \text{ kcal mol}^{-1}$. To emulate experimental reaction conditions, the partial pressures of H_2 and methanol (p_{H_2} and p_{MeOH}) were considered to be 19.34 and 8.78 atm at 423.15 K, respectively, and the G correction term was calculated accordingly⁸⁶. These thermodynamic parameter calculations were facilitated by the use of the Shermo code⁸⁷.

Data availability

Materials, general methods, instrumentation, synthetic protocols and DFT computational details are provided in the Supplementary Information PDF file. Crystallographic data for the structures reported in this Article have been deposited at the Cambridge Crystallographic Data Centre, under deposition numbers CCDC 2376631 ($\text{Co}_2\text{Cl}_2\text{BBTA}$), 2376626 ($\text{Co}_2(\text{OH})_2\text{BBTA}$), 2376634 ($\text{Co}_2(\text{SH})_2\text{BBTA}$), 2385609 ($\text{Ni}_2\text{Cl}_2\text{BBTA}$), 2385610 ($\text{Ni}_2(\text{OH})_2\text{BBTA}$), 2385611 ($\text{Ni}_2(\text{SH})_2\text{BBTA}$), 2376630 (Co-MFU-4l-Cl), 2376629 (Co-MFU-4l-OH), 2376628 (Co-MFU-4l-SH), 2376627 (Ni-MFU-4l-Cl), 2376633 (Ni-MFU-4l-OH) and 2376632 (Ni-MFU-4l-SH). Copies of the data can be obtained free of charge via <https://www.ccdc.cam.ac.uk/structures/>. The computational data supporting this study are available via Zenodo at <https://doi.org/10.5281/zenodo.15074051> (ref. 88). PXRD patterns, N_2 isotherms, XPS spectra and PDF data are available via Figshare at <https://doi.org/10.6084/m9.figshare.27327783> (ref. 89). Source data are provided with this paper.

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Author contributions

H.X., L.G. and O.K.F. conceived and supervised the project. H.X. designed and carried out the synthesis and catalytic experiments, and wrote the original draft together with M.M. M.A.K. contributed to catalysis methodology. M.M. performed theoretical calculations under the supervision of L.G. S.M.V., J.H. and L.M.T. carried out PDF measurements and analysis. K.F., D.A.G. and S.L. assisted with synthesis and materials preparation. S.S., S.R. and D.S. contributed to catalytic studies. K.M. participated in synthesis. X.W., F.S. and S.S. performed microscopy. J.M.N., K.M., W.G., J.G.V., Y.C., J.S.A. and K.O.K. contributed to the interpretation of results. J.M.N. also advised on catalysis. K.W.C. supervised PDF analysis. All authors discussed the results and contributed to revising the manuscript.

Competing interests

O.K.F. has financial interest in NuMat Technologies, a start-up company that is seeking to commercialize MOFs. The remaining authors declare no competing interests.

Additional information

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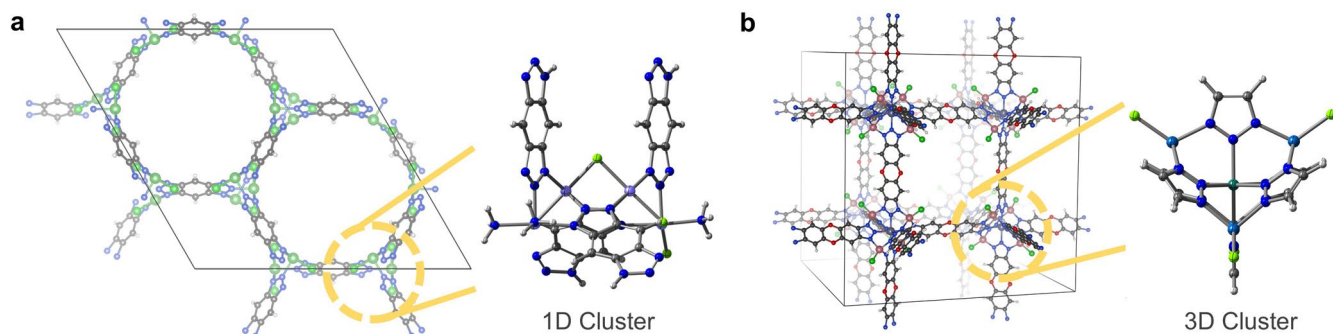
Extended Data Table 1 | Energy Decomposition Analysis of Ni^{II}-X Homolytic Dissociation in MFU-4l-Ni-X (X = Cl, OH, SH) at M06-D3(O)/def2-TZVP Level

Energy Contribution (kcal/mol)	MFU-4/-Ni-Cl	MFU-4/-Ni-OH	MFU-4/-Ni-S
Frozen (ΔE_{Fz})	13.3	23.7	4.7
Polarization (ΔE_{pol})	-15.0	-21.0	-16.4
Charge Transfer (ΔE_{CT})	-92.7	-85.9	-49.9
Total (ΔE_{tot})	-94.4	-83.2	-72.0

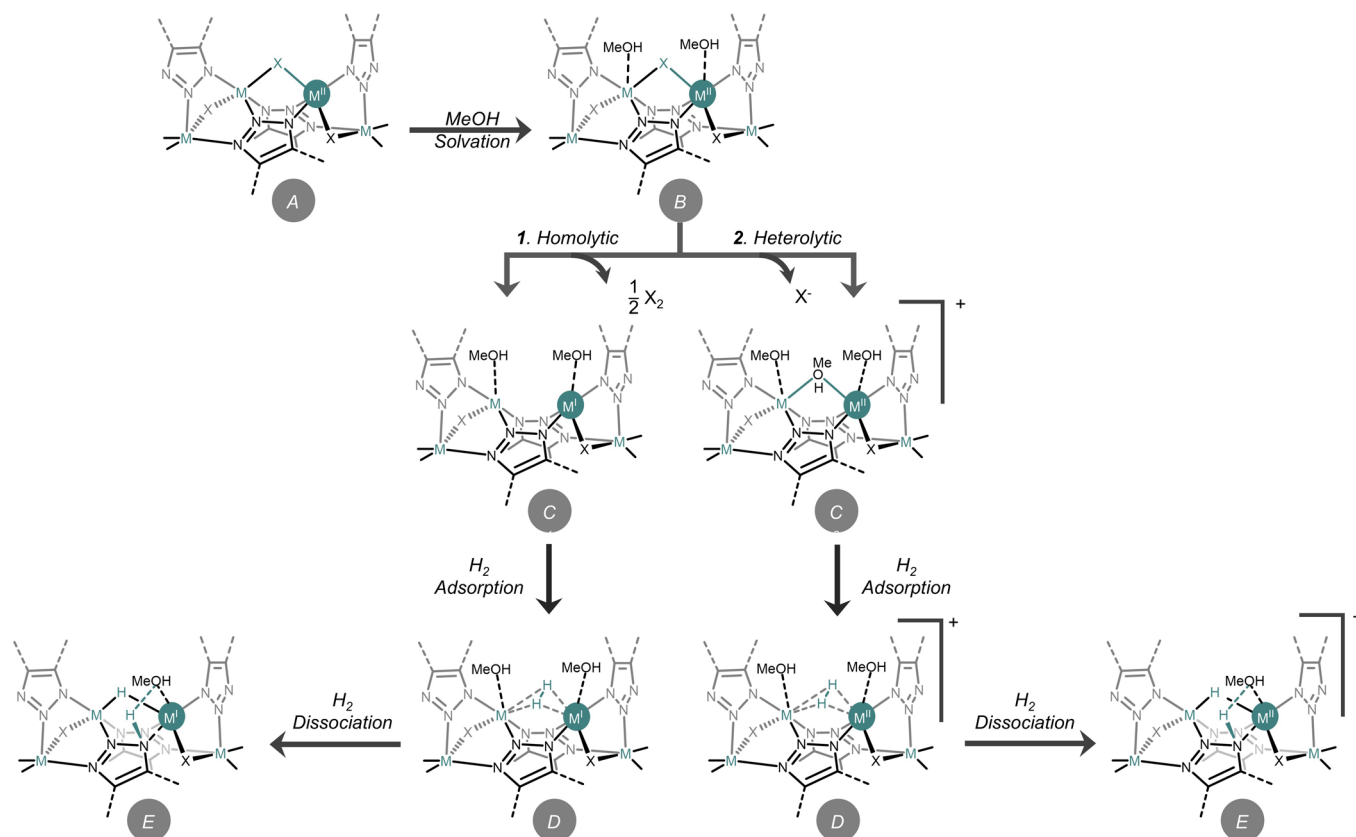
Extended Data Table 2 | Energy Decomposition Analysis of H₂ Binding at an M^I Site in both BBTA and MFU-4l Systems Calculated at the M06-D3(O)/def2-TZVP Level

Energy Contribution (kcal/mol)	BBTA-M-SH			
	Co	Ni	Co	Ni
Frozen ($\Delta E_{\text{Fiz}} = \Delta E_{\text{Fiz}} + \Delta E_{\text{Pauli}} + \Delta E_{\text{Disp}}$)	-3.0 (-8.9+13.5-7.6)	17.5 (-25.6 + 51.8 -8.8)	13.6 (-20.6 + 39.4 -5.1)	7.4 (-20.1 + 32.0 -4.4)
Polarization (ΔE_{pol})	-0.9	-11.7	-14.2	-13.0
Charge Transfer (ΔE_{CT})	-1.9	-9.6	-14.3	-15.2
Total (ΔE_{tot})	-5.8	-3.7	-14.9	-20.9

The individual components comprising ΔE_{Fiz} (= Electrostatic (ΔE_{elec}), Pauli (ΔE_{Pauli}), Dispersion (ΔE_{Disp})) are also shown.



Extended Data Fig. 1 | Periodic and cluster models used in theoretical studies of MOF catalysts. a, Structural models of M_2X_2BBTA . **b,** Structural models of $M-MFU-4l-X$. Both periodic and cluster models were used to represent the local coordination environments and extended structures (M = Co, Ni; X = Cl, OH, SH). Colour coding follows that used in Fig. 2.



Extended Data Fig. 2 | Schematic of open metal-site generation, hydrogen adsorption and dissociation for subsequent Ar-NO₂ reduction with M-BBTA-X catalysts. In the 1D BBTA system, MeOH initially coordinates to the square pyramidal metal centres, converting species **A** to **B**. Subsequent M-X bond cleavage (M = Co, Ni; X = Cl, OH, SH) can proceed either homolytically (**C1**) or heterolytically (**C2**), generating an open metal site for H₂ adsorption (**D1**, **D2**). The H-H bond then dissociates, facilitated by a N-atom on the ligand, forming species **E1** and **E2**. Although our computed results show relatively high energy requirements for the formation of active species **E**, our goal was to qualitatively understand trends related to MOF structure, metal node,

and X ligand. The cluster models employed frozen C atoms in the linkers to maintain structural rigidity of the MOF, potentially overestimating energetic barriers where remarkable ligand relaxation would occur, particularly during the H-H dissociation on the ligand (Conversion of species **D** to **E**). Future studies using fully relaxed *periodic* models of all proposed reaction intermediates should yield lower energy requirements for **E** formation. Additionally, while intrinsic barriers appear high due to solvent stabilization of species **B**, the apparent activation energy (E_{app}) observed experimentally can be lower when measured relative to separated reactants (Species **A**).